



Big Data System Design (MBAI 5110G)

Winter-2026

Week 1: Introduction

Lectured by:

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Outline

- Course Introduction
- Learning Outcomes
- Course Design Delivery
- Evaluation and Assessments
- Introducing Big Data Concept
- Sources and Different Forms of Big Data
- Big Data Infrastructure
- Big Data Life Cycle
- Big Data Technology and Tools
- Big Data Use Cases/Applications

Introduction: Brief Bio

Education and Research Experience

- **Postdoc & PhD:**
 - University of Toronto, Toronto, ON, Canada
 - York University, Toronto, ON, Canada
 - University of New Brunswick, Fredericton, NB, Canada
 - University of Canterbury, NZ

Research Interest

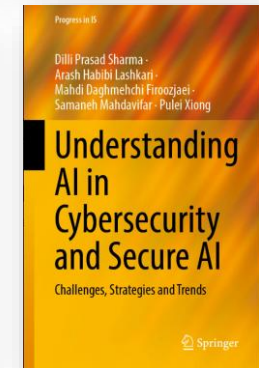
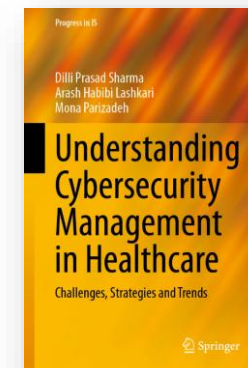
Artificial Intelligence, Cybersecurity, Privacy Preserving Methods, Big Data Analytics, Adversarial Machine Learning, Trustworthy and Responsible AI

Taught/Teaching Courses

Big Data Machine Learning Tools, Introduction, Artificial Intelligence, Deep Learning, AI Ethics and Data Governance, Principles of Information Assurance and Security, Cryptography, Network Security, Computer Security and Cyber Law, Computer Network, Operating Systems, Data Structure and Algorithms, Advanced Java Programming



Book Author/co-author



Cours Introduction

Learning Outcomes

- Assess diverse forms of big data, such as IoT data, social media data, and biometric data, to understand their characteristics and structure.
- Analyze and compare state-of-the-art big data storage architectures for scalability, performance, and governance in business contexts.
- Evaluate modern big data processing frameworks for data ingestion, cleaning, integration, real-time analytics, and large-scale retrospective analysis.
- Assess advanced analytics and artificial intelligence techniques and machine learning for extracting actionable business insights from big data
- Select appropriate big data storage, processing, and analytics solutions to meet specific organizational requirements, including event stream processing and real-time data pipelines.
- Design a big data system that integrates data models, big data storage, big data processing, and big data analytics, including artificial intelligence
- Assess cloud-based provisioning strategies for deploying and managing the big data system designs

Course Design and Delivery

- The course covers both **theoretical/analytical** and **practical** aspects
- **Hybrid delivery** format:
 - **Asynchronous** online learning
 - **In-person** lectures and discussions
- Required readings and active participation
- Course content is organized into weekly modules (see Course Outline)
- **Assessment Items:**
 - Assignments
 - Case Study
 - Projects
 - Presentation

Note: Please refer to the **Course Outline** for detailed information on course structure, assessment criteria, schedules, weekly topics, textbook/references, and reading materials

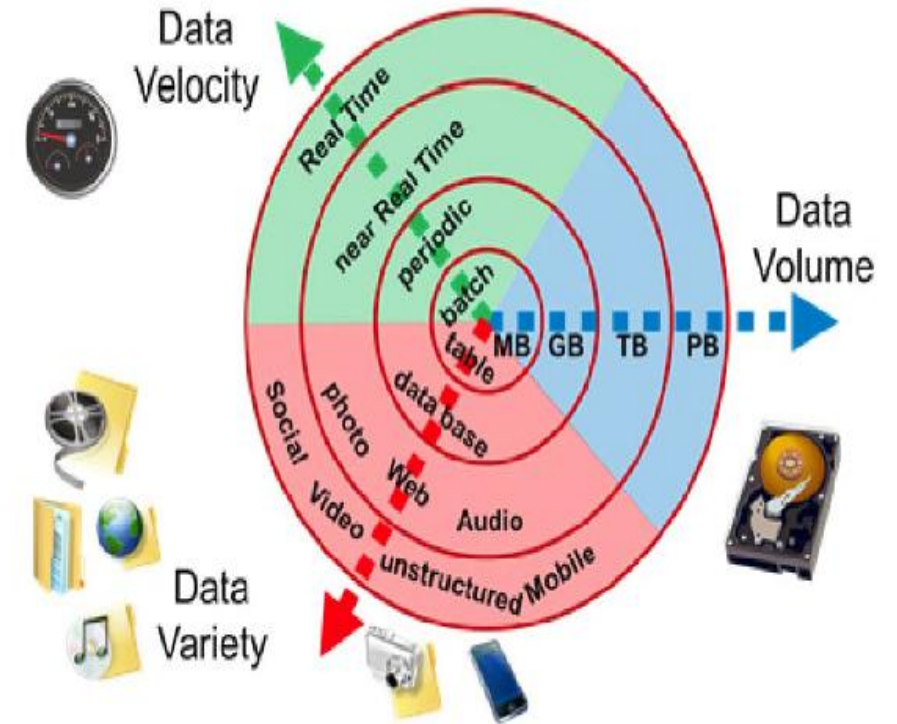
Evaluation & Assessment Items

Assessment Items	Weight	Individual/ Group
Assignment 1: Big Data Laboratory 1	10	• Individual • Deliverable: Report and Code
Assignment 2: Big Data Systems Design	10	• Individual • Deliverable: Design Documentation Report
Assignment 3: Big Data Laboratory 2	10	• Individual • Deliverable: Report and Code
Project 1: Big Data and Artificial Intelligence	20	• Individual-simulated workplace project • Deliverable: Project Report and Code
Case Study: Case Study Design, including problem statement, data cleaning, integration, data storage, real-time analytics, retrospective data analysis, and cloud provisioning	20	• Group work • Group size: 3 to 4 students • Deliverable: Case Study Report
Project 2: Prototype Development and Presentation	20	• Group work and Presentation • Group size: 3 to 4 students • Deliverable: Project Report, Code, and Presentation with Demo
Class Participation	10	
Total		100

Introduction to Big Data

What is Big Data?

- **Definition:**
 - ...extremely large data sets that may be analyzed computationally to **reveal patterns, trends, and associations**, especially relating to human behavior and interactions.
- **5Vs of Big Data:**
 - Initially 3 V's, namely **Volume, Velocity, and Variety**.
 - Then **Veracity** got added, making it 4 V's.
 - Then **Value** got added, making it 5V's.
 - Adding more Vs, now making it up to 13 V's!



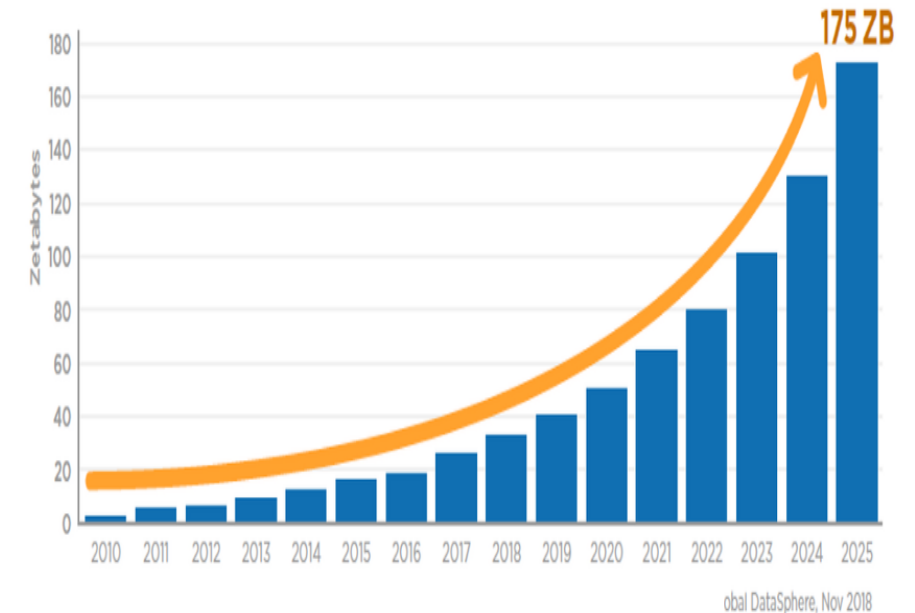
Introduction to Big Data (cont.)

- **Volume:**

- It refers to the size of Big Data.
- The rapidly increasing volume of data is due to cloud-computing traffic, IoT, mobile traffic, etc.

- **Velocity:**

- It refers to the speed at which the data is being accumulated. This is mainly due to IoTs, mobile data, social media, etc.

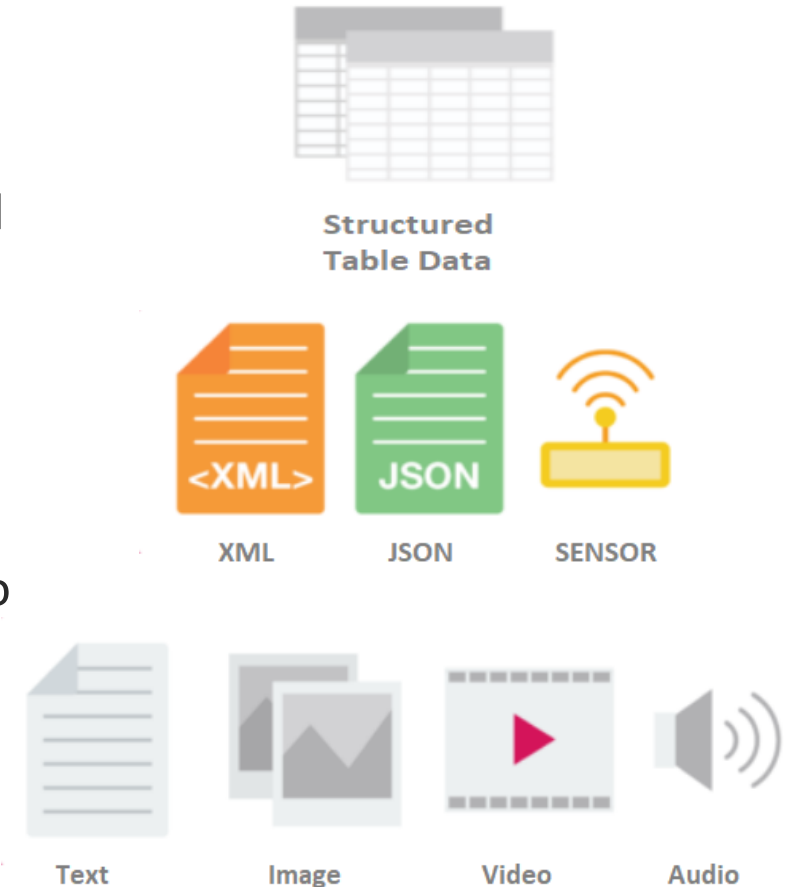


VELOCITY

Introduction to Big Data (cont.)

- **Variety:**

- It refers to **Structured**, **Semi-structured**, and **Unstructured** data due to different sources of data generated either by humans or by machines.
- **Structured data:** It's the traditional data that is organized and conforms to the formal structure of data.
 - This data can be stored in a relational database
- **Semi-structured data:** It's semi-organized data. It doesn't conform to the formal structure of data.
 - Example: Log files, JSON files, Sensor data, CSV files, etc.
- **Unstructured data:** It's not organized data and doesn't fit into row and column structure of a relational database.
 - Example: Text files, Emails, images, videos, voicemails, audio files etc.



Introduction to Big Data (cont.)

- **Veracity:**

- It refers to the assurance of **quality/integrity/credibility/accuracy** of the data.
- As data is collected from multiple sources, so we need to check the data for accuracy before using it for business insights.

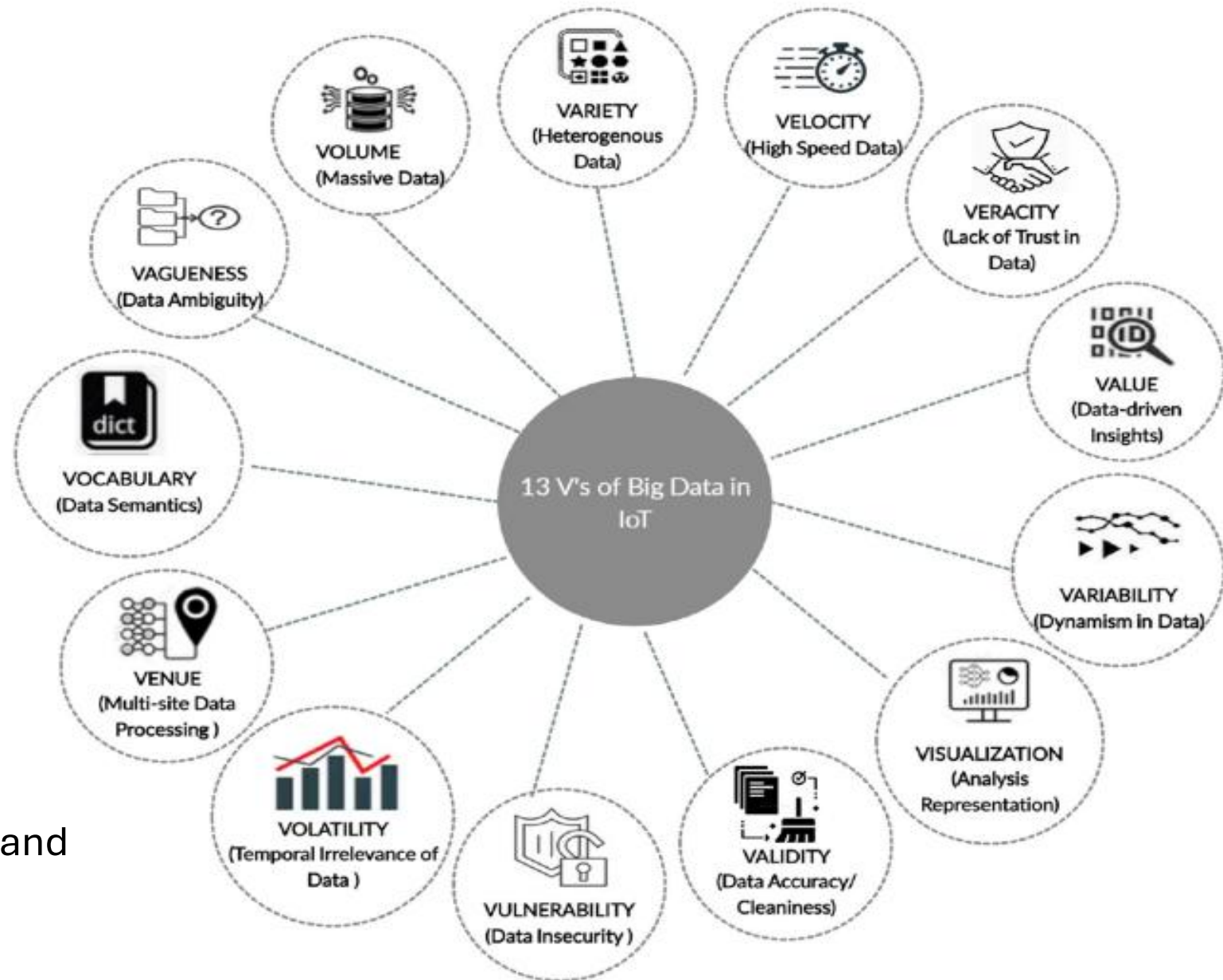


- **Value:**

- Just because we collected lots of Data, it's of no value unless we garner some insights out of it.
- Value refers to how useful the data is in decision making. We need to extract the value of the Big Data using proper analytics.



13 Vs



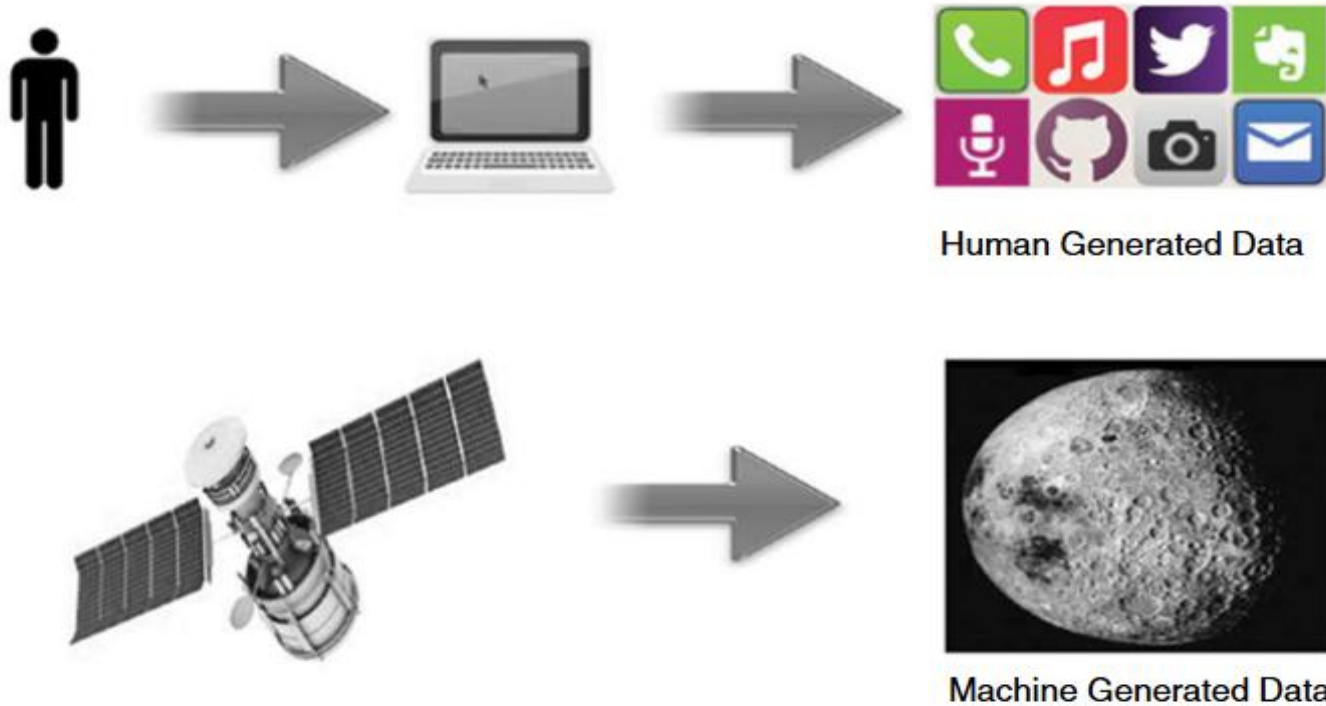
- Initially 3Vs, then 5Vs, and now 13Vs

Sources of Big Data

- **Sensors and IoT Devices:**
 - Smart city (pollution, traffic)
 - Wearable devices (heart-rate, steps)
 - Industrial machines (temperature, vibration),
 - smart home appliances (fridges)
- **Social Media:**
 - Twitter, LinkedIn, Facebook, Instagram, YouTube, posts, etc.
- **Healthcare systems:**
 - Demographics, diagnoses, medications, lab results, clinical notes, ECG, X-Ray, MRI, etc.
- **Web and Multimedia:**
 - Website contents
 - Clickstream (page view, clicks, session paths)
 - Search queries, Cookies, online purchases, Images, audios, videos, etc.
- **Business and Organization:**
 - ERP, CRM, HRM, SCM, Payroll, Accounting, Policy, etc.
- **Log and Events:**
 - System logs
 - Application logs,
 - Web server logs
 - Network traffic logs
 - Database server logs
 - User access logs, etc.



Different Types of Data



- Structured Data
- Unstructured Data
- Semi-structured Data

Structured Data

- Data that can be stored in a relational database in tabular format

Employee ID	Employee Name	Sex	Salary
334332	Daniel	Male	\$2300
334333	John	Male	\$2000
338332	Michael	Male	\$2800
339232	Diana	Female	\$1800
337891	Joseph	Male	\$3800
339876	Agnes	Female	\$4000

Semi-Structured Data

- Semi-structured data are those that have a structure but do not fit into a relational database.
- Semi-structured data are organized, which makes it easier to analyze when compared to unstructured data
 - JSON and XML are examples of semi-structured data representation

```
<?xml version = "1.0"?>
<Company>
  <Employee>
    <EmployeeId>339876</EmployeeId>
    <FirstName>Joseph</FirstName>
    <LastName>Agnes</LastName>
    <Sex>Female</Sex>
    <Salary>$4000</Salary>
  </Employee>
</Company>
```

Unstructured Data

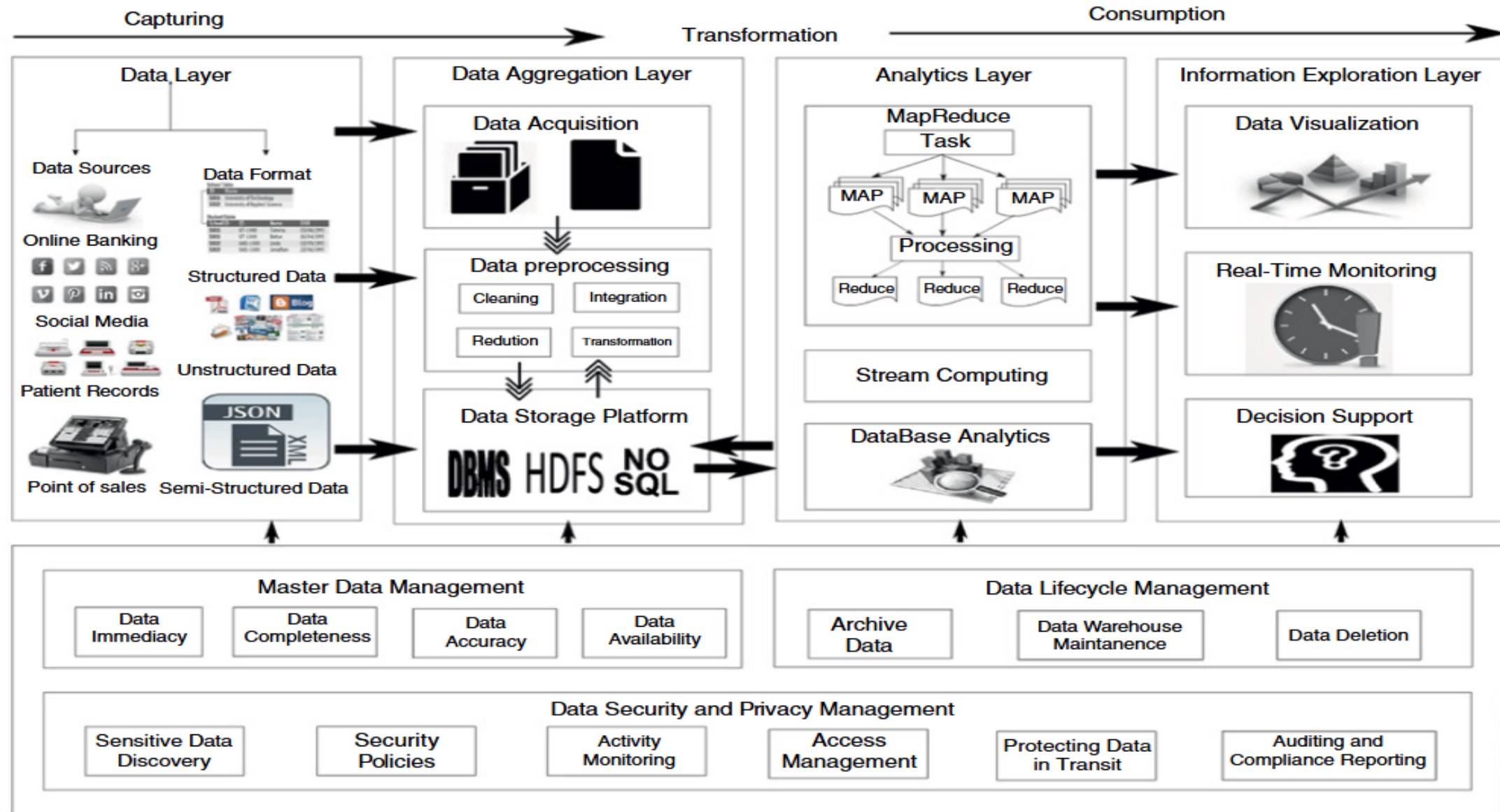
- Data that are raw, unorganized, and do not fit into the relational database systems are called unstructured data
 - video, audio, and social media posts, images, e-mails, text files
- Unstructured data usually resides on either text files or binary files
- Difficult to store and index
- Unstructured data lacks a predefined format but contains the richest information, making it central to Big Data analytics and artificial intelligence.



Big Data Infrastructure

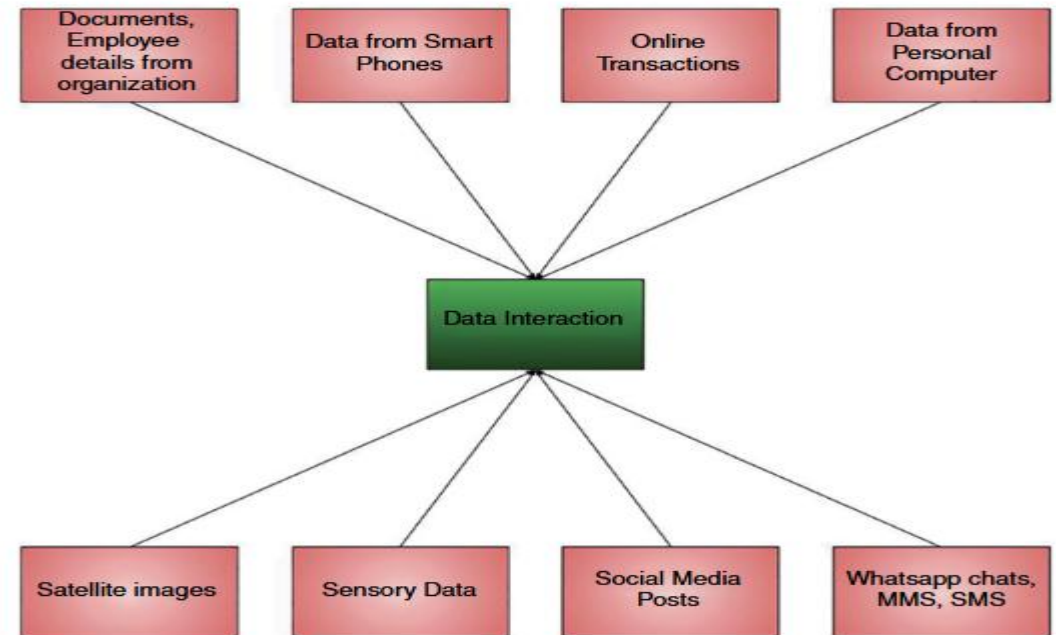
- The core components of big data technologies are the tools and technologies that provide the capacity to ***store, process, and analyze the data.***
- The key technologies include
 - **Hadoop:** a framework for processing large-scale datasets
 - **Hadoop Distributed File System (HDFS):** It is used to store large datasets with streaming access patterns running on low-cost commodity hardware
 - **MapReduce:** batch-processing programming model for the Hadoop framework, supports parallel and distributed computing environments

Big Data Life Cycle

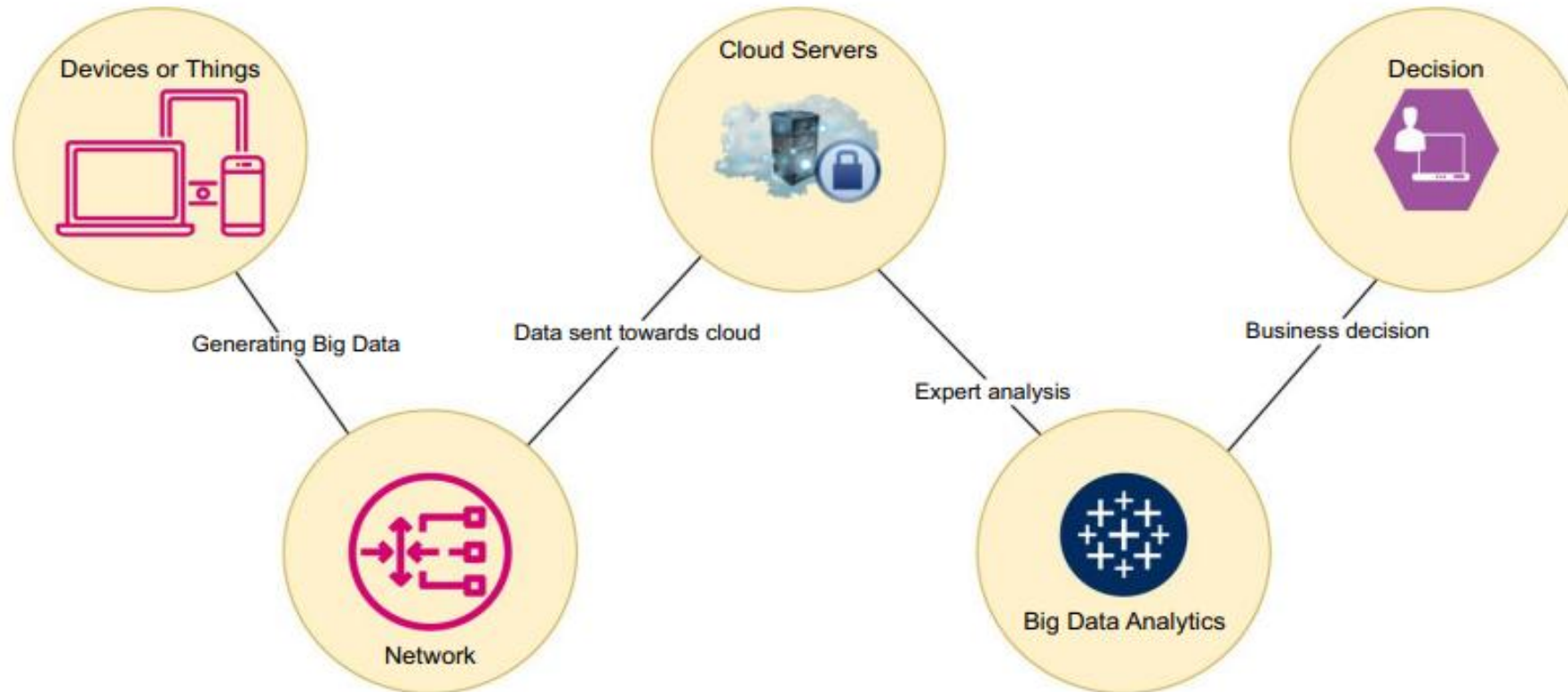


Phases of Big Data Life Cycle

- Big Data Generation
- Data Aggregation
- Data Preprocessing
 - Data integration
 - Data cleaning
 - Data reduction
 - Data transformation
- Big Data Analytics
- Visualizing Big Data

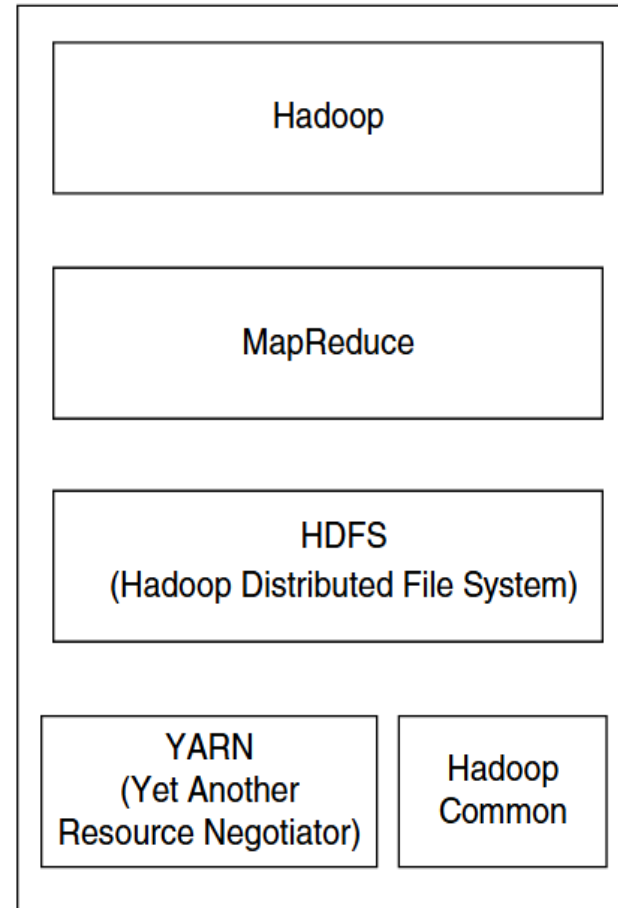


Big Data Application Workflow



Big Data Technology and Tools

- Hadoop
- Apache Spark
- NoSQL databases
- Data warehousing
- Machine learning
- Data integration tools
- Data visualization tools
- Cloud Computing

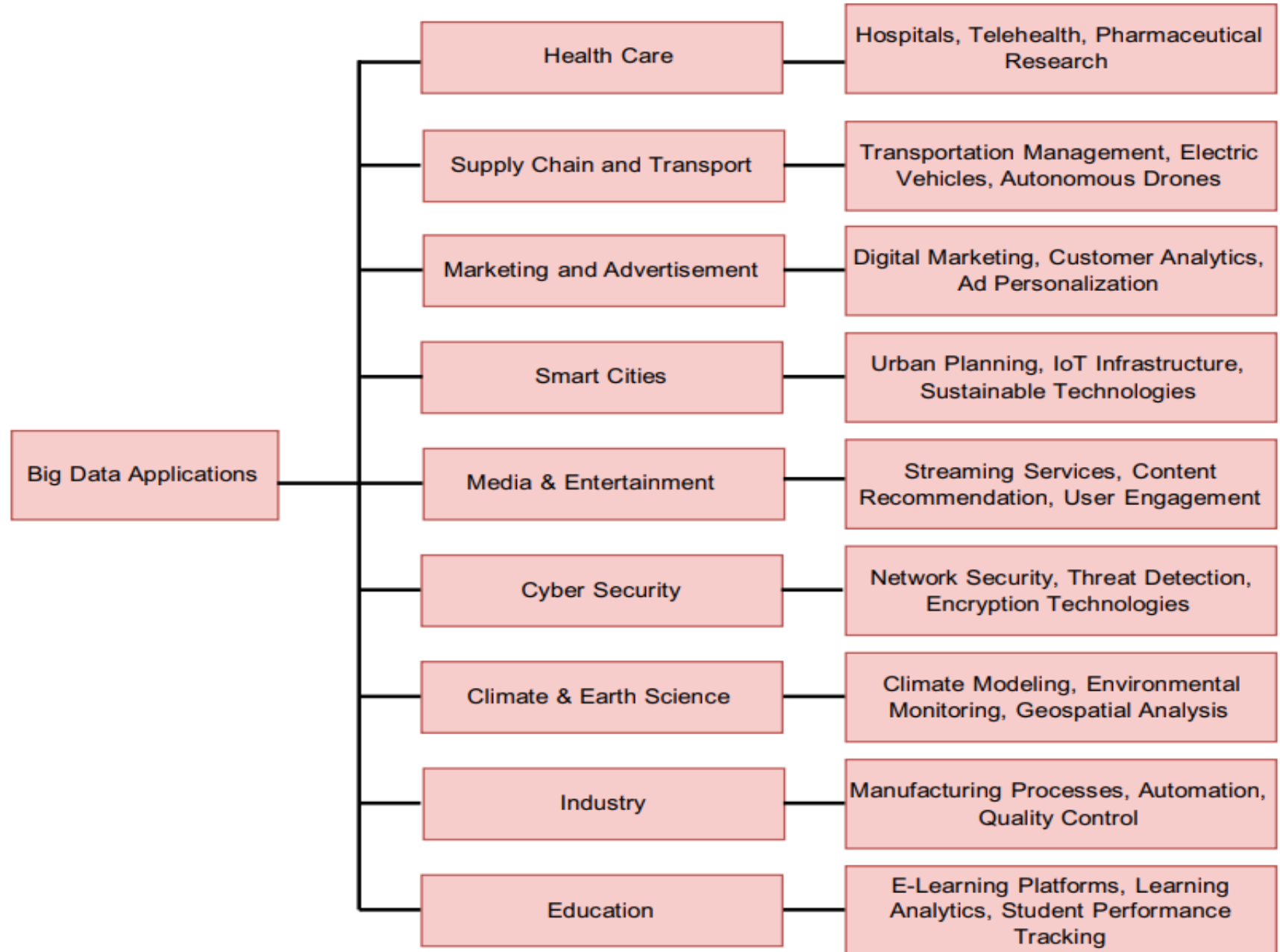


Hadoop core components



Common Big Data Analytics Tools

Big Data Application Classification



Readings & References

- [Chapter 1](#) (Textbook): Big Data: Concepts, Technology, and Architecture, Balamurugan Balusamy, Nandhini Abirami R, Seifedine Kadry, Amir H. Gandomi, Wiley, 2021
- Article: Badshah, A., Daud, A., Alharbey, R. et al. [Big data applications: overview, challenges and future](#). Artif Intell Rev 57, 290 (2024). <https://doi.org/10.1007/s10462-024-10938-5>
- Article: Maggi Bansal, Inderveer Chana, and Siobhán Clarke. 2020. [A Survey on IoT Big Data: Current Status, 13 V's Challenges, and Future Directions](#). ACM Comput. Surv. 53, 6, Article 131 (November 2021), 59 pages. <https://doi.org/10.1145/3419634>

Next Session: Big Data Storage Solutions

Q/A

Thank you!